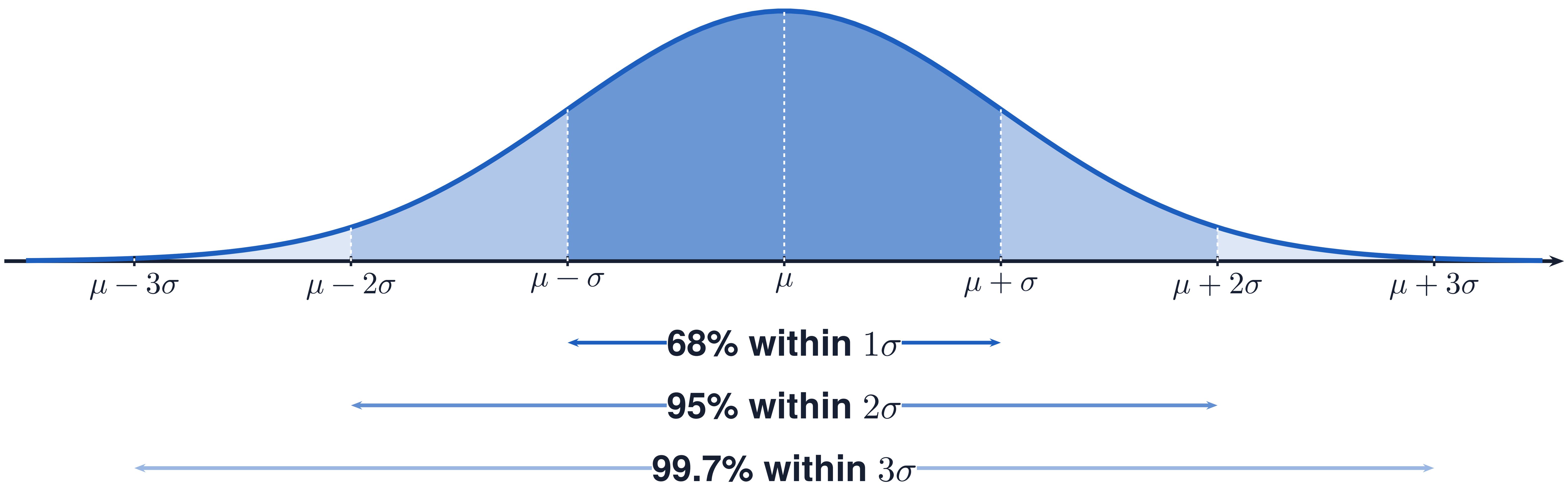
THE NORMAL DISTRIBUTION

new 2.11 • old 1.10

ZONE A · THE CURVE + THE FORMULA



$$z = \frac{x-\mu}{\sigma}$$

$x \longrightarrow z \longrightarrow$ area

normalcdf: value to probability

area $\longrightarrow z \longrightarrow x$

invNorm: probability to value

ZONE B · WHAT THE SYMBOLS MEAN

CENTER + SPREAD

$X \sim N(\mu, \sigma)$
μ: mean σ: standard deviation
Total area under the curve is 1.

STANDARDIZE THE DISTANCE

z counts SDs from the mean.
Positive: above μ. Negative: below μ.
At the mean, $z = 0$.

ZONE C · SAY IT IN WORDS

"________ is _____ standard deviations [above / below] the mean of _________________."

"About _____% of ____________ have ______________ below _______."

ZONE D · WATCH OUT

DRAW BEFORE YOU CALCULATE

Draw the curve. Mark the value.
Shade the requested region first.

PERCENTILE = AREA TO THE LEFT

Multiply the left-tail area by 100.
The 68–95–99.7 rule gives approximate percentages for a normal model.

see WS old u1_l10